

Branch Specialization in Ordinary Attention Heads

Toy Ontology v2: Structural Role Affinity, Specialization, and Modularity

Codex autonomous research checkpoint

2026-05-23

Checkpoint Purpose

- User direction: keep the unit as an ordinary attention head.
- Question: do different-shaped attention heads make some functional roles more likely to be learned by particular head types?
- This round expands from a small 6-row ontology to a 20-role ontology.
- Every result is shown as baseline, heterogeneous result, interpretation.

Strong current claim

Heterogeneous ordinary attention-head dimensions induce structural role affinity and increase role-level specialization in this toy setting.

Cautious current claim

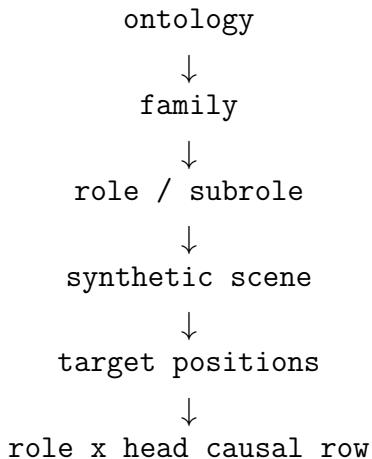
Functional modularity can improve in some layouts, but heterogeneity alone does not guarantee family-level modularity.

Three Questions

Question	What it asks	Metric here
Structural role affinity	Which structural head type gets a role?	Top dimension, largest-dim top rate, top-dim mass
Functional specialization	Is one role concentrated into one or a few heads?	Top role mass, effective heads
Functional modularity	Do related roles use similar head distributions?	Family gap and ARI

Specialization is role-level concentration. Modularity is family-level clustering. They can disagree.

Ontology Hierarchy



Toy Ontology v2

Family	Roles
copy_transport	local copy, previous token, KV lookup, duplicate token
induction	short induction, long induction, n-gram induction, false-induction control
position_boundary	BOS sink, SEP sink, fixed offset, punctuation boundary
suppression_conflict	distractor suppression, anti-copy, wrong-key suppression, recency conflict
entity_coreference	repeated-name detection, pronoun antecedent, IOI name mover, negative-name control

Total: 5 families x 4 roles = 20 measured role rows.

Example Scenes

Role	Scene	Target
local copy	[x, LOCAL_SEP, x]	At separator, predict the earlier token.
KV lookup	[KV_SEP, k1,v1,...,kq,vq]	At query key, predict paired value.
induction	[IND, x1..x6, x1..x6]	On second copy, predict continuation from first copy.
recency conflict	[key, old, key, recent, key, old]	Prefer older value over closer recent value.
IOI name mover	[A,B,A,IOI_QUERY,B]	Predict non-repeated name.

Full 20-role examples are in `data/role_dataset_examples.json`.

Literature Anchors

Source	Venue / year	Roles motivated
Clark et al., BERT attention Voita et al., specialized heads	BlackboxNLP at ACL 2019 ACL 2019	delimiter, fixed position, syntax, coreference specialized, positional, syntactic, rare-token heads
Elhage et al., Transformer Circuits Olsson et al., induction heads	Transformer Circuits 2021 Transformer Circuits / arXiv 2022	QK/OV, previous-token heads, circuits induction and previous-token composition
Wang et al., IOI circuit	ICLR 2023	name movers, negative heads, duplicate- token roles
McDougall et al., copy suppression	arXiv/OpenReview 2023; BlackboxNLP 2024	copy suppression and negative copy effects

Experiment Settings

Item	Value
Unit	ordinary attention head
Model	tiny decoder-only transformer
Dataset	one synthetic 20-role next-token sequence generator
Total attention dimension	128 in every config
Seeds	1, 2, 3, 4, 5
Steps	1600
Eval examples	512
Measurement	single-head ablation, role x head causal matrix

Config	Head dims	Purpose
uniform4	[32,32,32,32]	four-head uniform baseline
uniform2	[64,64]	strong fewer/wider uniform baseline
hetero4_unique_mild	[16,24,40,48]	mild four-head heterogeneity
hetero4_unique_64	[8,16,40,64]	one 64-dim structural type
hetero4_unique_extreme	[8,16,24,80]	extreme four-head imbalance
hetero2_unique_mild	[48,80]	mild two-head heterogeneity
hetero2_unique_mid	[32,96]	medium two-head heterogeneity
hetero2_unique_extreme	[16,112]	extreme two-head heterogeneity

Learnability Check

Config	Mean role accuracy	Min role accuracy
uniform4	1.000	0.998
uniform2	0.999	0.991
hetero4_unique_mild	1.000	0.999
hetero4_unique_64	1.000	0.999
hetero4_unique_extreme	1.000	0.999
hetero2_unique_mild	1.000	0.999
hetero2_unique_mid	1.000	0.996
hetero2_unique_extreme	1.000	0.997

Interpretation: comparisons are not driven by failed training.

Q1: Structural Role Affinity

Hetero config	Chance largest top	Observed largest top	Mean top-dim mass
[16,24,40,48]	0.25	0.50	0.759
[8,16,40,64]	0.25	0.65	0.821
[8,16,24,80]	0.25	0.82	0.912
[48,80]	0.50	0.66	0.839
[32,96]	0.50	0.82	0.871
[16,112]	0.50	0.94	0.937

Interpretation: larger structural head types attract roles far above chance, and the effect strengthens with imbalance.

Q2: Functional Specialization

Config	Specialization	Effective heads
uniform4	0.733	2.400
uniform2	0.684	2.166
hetero4_unique_mild	0.726	2.206
hetero4_unique_64	0.776	1.906
hetero4_unique_extreme	0.849	1.522
hetero2_unique_mild	0.782	1.802
hetero2_unique_mid	0.776	1.692
hetero2_unique_extreme	0.780	1.636

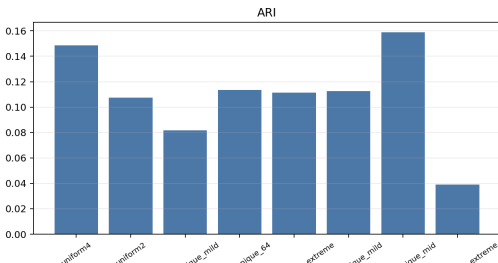
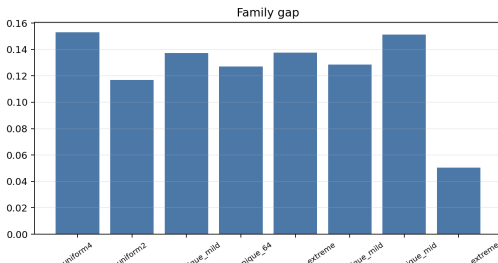
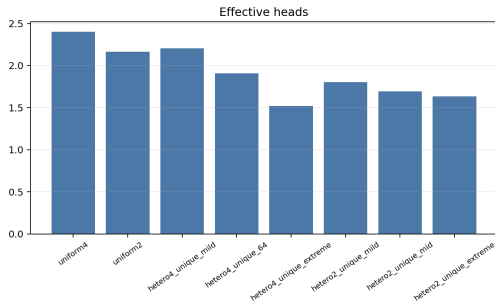
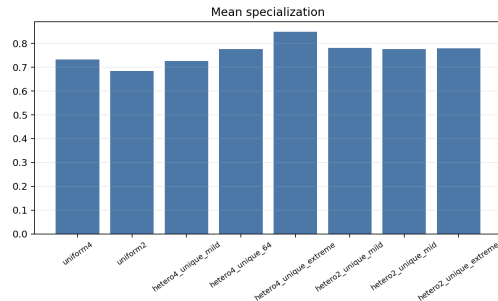
Interpretation: heterogeneity generally concentrates each role into fewer causal heads.

Q3: Functional Modularity

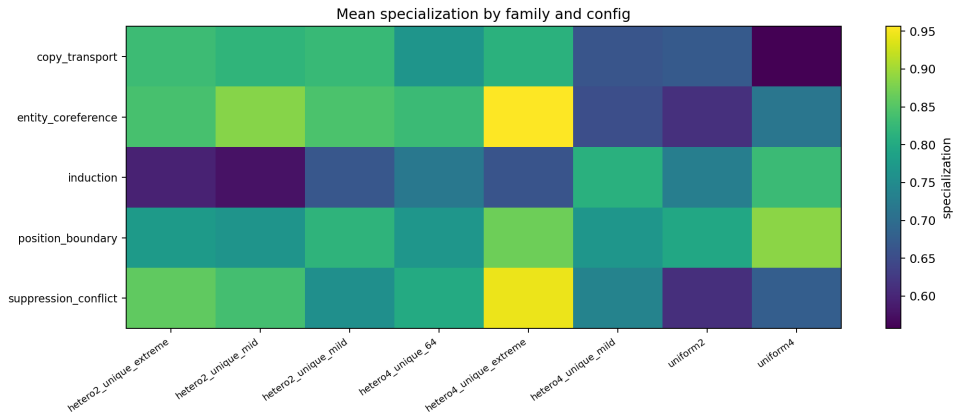
Config	Family gap	ARI
uniform4	0.153	0.149
uniform2	0.117	0.108
hetero4_unique_mild	0.137	0.082
hetero4_unique_64	0.127	0.113
hetero4_unique_extreme	0.138	0.111
hetero2_unique_mild	0.128	0.112
hetero2_unique_mid	0.151	0.159
hetero2_unique_extreme	0.050	0.039

Interpretation: modularity is not automatic. The best two-head hetero layout is promising, but extreme imbalance can collapse family structure.

Aggregate Metric Plot



Family Specialization Heatmap



The key check: if a role only likes head index H3, moving the 64-dim head should not move the role.

Layouts tested:

- [8,16,40,64]
- [64,8,16,40]
- [8,64,16,40]
- [8,16,64,40]

Layout Results

Layout	64 position	Largest-top rate	Family gap	ARI
[8,16,40,64]	H3	0.65	0.127	0.113
[64,8,16,40]	H0	0.68	0.100	0.089
[8,64,16,40]	H1	0.67	0.086	0.096
[8,16,64,40]	H2	0.60	0.186	0.189

Interpretation: the 64-dim preference persists when the 64-dim head moves, but family modularity is layout-sensitive.

Which Roles Followed The 64-Dim Type?

Role	64-dim top cases	Rate
local_copy	20/20	1.00
negative_name_control	19/20	0.95
recency_conflict	19/20	0.95
wrong_key_suppression	19/20	0.95
pronoun_antecedent	18/20	0.90
kv_lookup	17/20	0.85
simple_ioi_name_mover	17/20	0.85

Some induction and boundary roles were less 64-locked. This is evidence for role-specific affinity, not just universal collapse.

Claim	Status	Reason
Heterogeneous dimensions create structural role affinity	Strong toy evidence	Above-chance largest-head selection and layout controls
Heterogeneous dimensions increase specialization	Strong toy evidence	Higher top-role mass and fewer effective heads
Heterogeneous dimensions create modularity	Mixed	Some layouts improve, others do not; extreme imbalance hurts
Toy ontology maps to real LM roles	Not proven yet	Literature-grounded roles, but synthetic dataset

Continue the project.

The evidence is strong enough for the structural-affinity and specialization story.

The modularity story should be treated as a separate research question, not as something automatically implied by heterogeneous dimensions.

Next Steps

- 1 Extend layout controls for the best modularity layout [8,16,64,40].
- 2 Add more induction and position-boundary subroles because they behave differently from copy/conflict roles.
- 3 Run real-model validation with ordinary pretrained heads: local copy, induction, delimiter/BOS, copy suppression, and IOI-style probes.
- 4 Add bootstrap confidence intervals or permutation tests before paper drafting.
- 5 Keep future reporting format: baseline result, heterogeneous result, interpretation.

Provenance

Main memo:

`doc/experiments/phase3/phase3_toy_role_ontology_v2.md`

Main result root:

`results/phase3_toy_role_ontology_v2_full_1600_20260523`

Layout control root:

`results/phase3_toy_role_ontology_v2_layout_1600_20260523`

Scripts:

`scripts/toy_role_ontology_v2_head_dim_intervention.py`

`scripts/analyze_role_ontology_v2.py`

Reproduction Commands

```
python scripts/toy_role_ontology_v2_head_dim_intervention.py \  
  --role-set v2_full \  
  --configs uniform4 uniform2 hetero4_unique_mild hetero4_unique_64 \  
    hetero4_unique_extreme hetero2_unique_mild hetero2_unique_mid \  
    hetero2_unique_extreme \  
  --seeds 1 2 3 4 5 --steps 1600 --batch-size 128 \  
  --eval-examples 512 \  
  --output-dir results/phase3_toy_role_ontology_v2_full_1600_20260523  
  
python scripts/analyze_role_ontology_v2.py \  
  results/phase3_toy_role_ontology_v2_full_1600_20260523 \  
  --title "Toy Ontology v2 Full 20-Role Sweep"
```